Spatial–Spectral 1DSwin Transformer With Groupwise Feature Tokenization for Hyperspectral Image Classification

Yifei Xu, Yixuan Xie, Bicheng Li, Chuanqi Xie, Yongchuan Zhang, Aichen Wang, and Li Zhu

***Abstract*— The hyperspectral image (HSI) classification aims to assign each pixel to a land-cover category. It is receiving increasing attention from both industry and academia. The main challenge lies in capturing reliable and informative spatial and spectral dependencies concealed in the HSI for each class. To address the challenge, we propose a spatial–spectral 1DSwin (SS1DSwin) Transformer with groupwise feature tokenization for HSI classification. Specifically, we reveal local and hierarchical spatial–spectral relationships from two different perspectives. It mainly consists of a groupwise feature tokenization module (GFTM) and a 1DSwin Transformer with cross-block normalized connection module (TCNCM). For GFTM, we reorganize an image patch into overlapping cubes and further generate groupwise token embeddings with multihead self-attention (MSA) to learn the local spatial–spectral relationship along the spatial dimension. For TCNCM, we adopt the shifted windowing strategy when acquiring the hierarchical spatial–spectral relationship along the spectral dimension with 1-D window-based MSA (1DW-MSA) and 1-D shifted window-based MSA (1DSW-MSA) and leverage cross-block normalized connection (CNC) to adaptively fuse the feature maps from different blocks. In SS1DSwin, we apply these two modules in order and predict the class label for each pixel. To test the effectiveness of the proposed method, extensive experiments are conducted on four HSI datasets, and the results indicate that SS1DSwin outperforms several current state-of-the-art methods. The source code of the proposed method is available at https://github.com/Minato252/SS1DSwin.**

***Index Terms*— 1-D shifted window-based multihead self-attention (1DSW-MSA), 1-D window-based MSA (1DW-MSA), cross-block normalized connection (CNC), hyperspectral image (HSI) classification, Swin Transformer.**

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NOMENCLATURE

$\mathbf{I}$	Given input hyperspectral image (HSI).
$\mathbf{I}_c$	Image matrix of $\mathbf{I}$ at the cth channel.
$\mathbf{I}^n$	Normalized $\mathbf{I}$.
$\mathbf{I}_c^n$	Image matrix of $\mathbf{I}^n$ at the cth channel.
$\mathbf{I}^{\text{pn}}$	Normalized and padded $\mathbf{I}$.
$\mathbf{P}$	Image patch of the given pixel cropped from $\mathbf{I}^{\text{pn}}$.
$\mathbf{P}_c$	Image matrix of $\mathbf{P}$ at the cth channel.
$\mathbf{P}^{\text{pad}}$	Padded $\mathbf{P}$.
$\mathbf{P}_c^{\text{pad}}$	Image matrix of $\mathbf{P}^{\text{pad}}$ at the cth channel.
$\mathbf{G}$	Set of group cubes.
$\mathbf{G}_c$	cth group cube of $\mathbf{G}$.
$\mathbf{X}_c$	Reshaped $\mathbf{G_c}$.
$\mathbf{PE^1}$	Learnable positional embedding in token generation module (TGM).
$\mathbf{X}'_c$	Sum of $\mathbf{X}_c$ and $\mathbf{PE^1}$.
$\mathbf{Z}_c$	Result of applying multihead self-attention (MSA) on $\mathbf{X}_c$.
$\mathbf{T}$	Matrix formed by token embeddings.
$\mathbf{T}_c$	cth token embedding of $\mathbf{T}$.
$\mathbf{PE^2}$	Learnable positional embedding in 1DSwin Transformer with cross-block normalized connection module (TCNCM).
$\mathbf{F}_{\text{TM}}^{s_1}$	Sum of $\mathbf{T}$ and $\mathbf{PE^2}$.
$\mathbf{F}_{\text{TM}}^{s_i}$	Result of token merging layer in stage $i, i \neq 1$.
$\mathbf{F}_{b_j}^{s_i}$	Result of the jth 1DSwin Transformer block in stage i.
$\mathbf{F}_{b_{\max_i}}^{s_i}$	Result of the last 1DSwin Transformer block in stage i.
$\hat{F}_{b_j}^{s_i}$	Result of 1-D window-based MSA (1DW-MSA) in the jth 1DSwin Transformer block in stage i.
$\mathbf{F}_{\text{CNC}}^{s_i}$	Result of cross-block normalized connection (CNC) in stage i.
$\mathbf{F}_{\text{CNC}_m}^{s_i}$	Image matrix of $\mathbf{F}_{\text{CNC}}^{s_i}$ at the mth channel.
$\mathbf{F}_{\text{TM}}^{s'_i}$	Result of the concatenated normalization layer corresponding to $\mathbf{F}_{\text{TM}}^{s_i}$.
$\mathbf{F}_{b_{\max_i}}^{s'_i}$	Result of the concatenated normalization layer corresponding to $\mathbf{F}_{b_{\max_i}}^{s_i}$.
$\mathbf{F}^{\text{gap}}$	Final output of TCNCM.
$\mathbf{Y}$	Predicted label of the given pixel.

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I. Introduction

A HYPERSPECTRAL imaging technique is of great significance in the field of remote sensing. Each pixel in HSI is a 1-D data sequence containing hundreds of narrow bands corresponding to wavelengths from visible light to near-infrared light. The abundant spatial and spectral information captured by hyperspectral sensors has been widely used for various practical applications [1], [2], [3], [4]. As a critical step, HSI classification lays the foundation for these applications, assigning a land-cover category to each pixel [5].

In the early stages, how to extract informative spatial–spectral features has been explored in traditional HSI classifiers. Various techniques involving support vector machine (SVM) [6], K-nearest neighbor (KNN) [7], random forest (RF) [8], and their variants [9], [10], [11] are used directly to predict the label for each pixel. However, their lower generalization ability often results in unsatisfactory performance when the training set becomes larger and more complex. Along with the emergence of deep learning, there are kinds of remarkable research works on HSI classification, such as convolutional neural networks (CNNs) [12], [13], [14], [15], [16], [17], [18], [19], deep belief networks (DBNs) [20], [21], [22], recurrent neural networks (RNNs) [23], capsule networks [24], [25], generative adversarial networks (GANs) [26], [27], and graph convolutional networks (GCNs) [28], [29]. Although these deep learning-based methods have great success in producing promising classification results, they always have their own inevitable drawbacks in revealing the spatial–spectral relationship in HSI. First, CNNs are overly concerned with spatial information, thereby their ability to capture the spatial–spectral relationship along the spectral dimension remains inadequate. Second, with regard to DBNs, they are struggling to address the problems of well-known coadaptation of latent factors [30], [31] and a large scale of "dead" (never responding) or "potential over-tolerant" (always responding) latent neurons [22]. Third, when RNNs strive to capture the spatial–spectral relationship from long-range sequence data, they suffer from gradient vanishing and extreme dependence on the order of spectral bands. Fourth, capsule networks work slowly in the inference stage due to the high computational complexity and numerical instability of iterative routing mechanisms, and their complex routing procedures may decrease their expressivity [24]. Fifth, while GANs are powerful to empower unsupervised and weakly supervised HSI classifiers, they always encounter the failures of model collapse and non-convergence. Sixth, as an intrinsic defect, GCNs are beset by over-fitting [32] and over-smoothing [33] when the number of layers is greater than three.

Very recently, with the prevalence of Transformer [34] networks in the field of computer vision, a group of Transformer-based methods have been applied to HSI classification [35], [36], [37], [38], [39], [40], [41]. As a core module of pure Transformers, global MSA [34] allows the attention function to extract information from different representation subspaces. As is known to all, the weak inductive bias and capture of long-range dependencies in global MSA have been conceived to be beneficial to predictive performance. Nevertheless, the recent study [42] suggests that the appropriate constraints on MSA actually assist in learning strong representations, and the capture of long-range dependencies becomes a burden to model optimization. Besides, these Transformer-based methods are also faced with quadratic computation complexities, hindering their deployment in practical applications [43]. Later, borrowing the idea from Swin Transformer [43], 3-DSwinT-based hierarchical contrastive learning (3DSwinT-HCL) accepts the whole HSI as input and employs a local multiscale contrastive learning module to exploit multiscale semantic representations [44]. Unfortunately, it is unpractical to apply it due to the huge pretraining cost and complex 3-D Transformer blocks. In conclusion, it is challenging for Transformer-based methods to capture more reliable and informative spatial–spectral representations concealed in HSI to achieve robust classification efficiently.

To overcome these aforementioned issues, we rethink the feature extraction process on the HSI data from two different perspectives and design a novel Transformer-based HSI classification method termed spatial–spectral 1DSwin (SS1DSwin) Transformer with groupwise feature tokenization. Specifically, it characterizes local and hierarchical spatial–spectral relationships along the spatial and spectral dimensions, respectively. As illustrated in Fig. 1, SS1DSwin is mainly made of groupwise feature tokenization module (GFTM) and TCNCM. The GFTM regroups the input image patch into overlapping cubes and captures the local spatial–spectral relationship by individually projecting each cube into the token embedding through MSA along the spatial dimension. The TCNCM arranges to extract the hierarchical spatial–spectral relationship from these token embeddings via 1DW-MSA and 1-D shifted window-based MSA (1DSW-MSA) and employs CNC for fusing features across a group of 1DSwin Transformer blocks. In general, the main contributions of our work can be summarized as follows.

1) We propose a novel Transformer-based method named SS1DSwin that fully exploits the spatial–spectral relationship in HSI. To the best of our knowledge, it is the first attempt to explore the potential of Swin Transformer by reconstructing it in 1-D space for patchwise HSI classification.
2) We propose two effective modules GFTM and TCNCM to learn local and hierarchical spatial–spectral relationships from two perspectives. In addition, we design CNC in TCNCM to fuse features adaptively across 1DSwin Transformer blocks to further enhance the classification performance.
3) We conduct extensive experiments on four benchmark datasets (Indian Pines, Pavia University, Houston2013, and Salinas Scene). The results indicate that our proposed method outperforms the other state-of-the-art methods.

The remainder of this article is organized as follows. Section II discusses the traditional methods, classic deep learning-based methods, and Transformer-based methods used for HSI classification. In Section III, we detail the proposed SS1DSwin. Section IV gives the data description, experimental setup, results and discussion, computational complexity

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analysis, and ablation studies. Section V concludes this article in a comprehensive way.

II. Related Works

A. Traditional Methods

Before the prevalence of deep learning, the HSI classification approaches designed a series of handcrafted features to capture spectral information followed by a classifier at each pixel. KNN used the distance of the sample pixels to determine the label of a pixel [7]. Nevertheless, it encounters the difficulty of a "curse of dimensionality" caused by high-dimensional data. Following that, Ham et al. [8] used RF to achieve better performance. It applied a binary hierarchical multiclassifier to improve classifier generalization in HSI data processing. Benefiting from the superior ability of dimension reduction, SVM has been popular in processing HSI classification tasks [6]. However, the aforementioned methods do not have enough generalization ability to cope with different HSI data and require a lot of human effort and experience.

B. Classic Deep Learning-Based Methods

The recent success of deep learning has led to a surge of deep learning-based HSI classifiers. These works have used CNNs [12], [13], [14], [15], [16], [17], [18], DBNs [20], [21], [22], RNNs [23], [45], capsule networks [24], [25], GANs [46], [47], [48], and GCNs [28], [29] to extract features from HSI data. Among these models, Rasti et al. [13] applied 1-D CNN to assign a label to each pixel. However, it is limited by noise without considering spatial information. Inspired by CNN in the field of natural images, 2-D CNN was transplanted to explicitly model the spatial–spectral relationship in image patches [14]. However, its fatal defect lies in the utilization of an extra dimension reduction of the original HSI in spectral dimension. Later, several works were done using 3-D CNN to describe spatial–spectral information [16], [49]. Although 3-D CNNs achieve promising results, it is still not suitable for real-world applications due to the huge computational burden. To take advantage of 3-D CNN and mitigate the computation cost, 2-D and 3-D convolution worked together to enable a joint spatial and spectral information process [15], [19]. Attention-based adaptive spectral–spatial Kernel improved residual network (A2S2K-ResNet) [17] presented an attention-based adaptive spectral–spatial kernel improved residual network with spectral attention to capture discriminative spectral–spatial features for HSI classification. In order to simultaneously use pixels' spatial information and spectral information for HSI classification, Yang et al. [18] developed a new cross-mixing residual network made up of a 3-D residual structure, a 2-D residual structure, and a structure of assisted feature extraction. Chen et al. [21] combined DBN and conjugate gradient (CG) update algorithm to capture spectral features and pixel-centric spectral block features for HSI classification. Taking into account the spectral signature as a sequence, the cascaded RNN model adopted gated recurrent units (GRUs) to construct redundant and complementary information from HSI [45]. To capture the spatial relationship between different features effectively, a multiple attention-guided CapsNet was proposed to identify the most representative visual parts of the HSI through a detailed feature extractor coupled with attention mechanisms [25]. Bai et al. [46] combined GAN, Transformer encoder, and convolution block in a unified framework and achieved promising HSI classification results on a few labeled samples. After investigating CNNs and GCNs in the HSI classification, Hong et al. [29] developed a miniGCN to train large GCNs in a minibatch manner, which could accurately recognize out-of-sample data without retraining. Despite their respective advantages in learning spectral representations, almost all of them are incompetent in modeling sequential information effectively.

C. Transformer-Based Methods

Inspired by the success of Transformers in computer vision (CV) [34], [43], [50], several Transformer-based methods [35], [36], [37], [39], [40], [41] have been developed for HSI classification. Owing to the excellent ability to handle sequential data, Transformers were purely applied to HSI classification termed SpectralFormer (SF), which regarded spectral signals as sequence data and fed them into a group-wise spectral embedding module and cross-layer adaptive fusion for better discriminative performance [36]. Later on, by embedding convolution operations in the Transformer structure, hyperspectral image Transformer (HiT) captured subtle spectral discrepancies and transmitted local spatial context information in the HSI classification [37]. Sun et al. [38] constructed a spatial–spectral feature tokenization Transformer (SSFTT) to capture spatial–spectral features and high-level semantic features. Very recently, to overcome the demerits caused by MSA, Mei et al. [39] exhibited a group-aware hierarchical Transformer (GAHT) that confined MSA to the local spatial–spectral context and constructed the Transformer in a hierarchical manner. With full consideration of the image redundancy property, the merged image-spectrum structure, and the spatial prior knowledge of HSIs, local-enhanced spatial–spectral Transformer (LF) was designed based on the HSI2Token module and the local-enhanced Transformer encoder to strengthen the ability to identify pixels in HSI [40]. To make full use of multimodal data [e.g., light detection and ranging (LiDAR), multispectral image (MSI), synthetic-aperture radar (SAR), and digital surface model (DSM)], a new multimodal fusion Transformer (MFT) network [41] is proposed to achieve better generalization by efficiently fusing the information from HSI patch tokens and existing classification tokens. It should be noted that several above works [35], [36], [38], [39], [40] just translate pure Transformers from natural language processing (NLP) or CV to HSI directly. As discussed in work [44], significant differences, such as the size and the number of basic elements between NLP and HSI, are not been fully considered when applying NLP Transformers [35]. Despite the fact that visual Transformers [38], [39], [40], [41] show great potential in HSI classification, they still face quadratic computation complexities [43]. According to this study [42], their capture of long-range dependencies even hinders network optimization. The most similar to our work is 3DSwinT-HCL [44], which also borrowed the idea from Swin Transformer. Unlike the pixel-by-pixel HSI classification, 3DSwinT-HCL only accepted the whole

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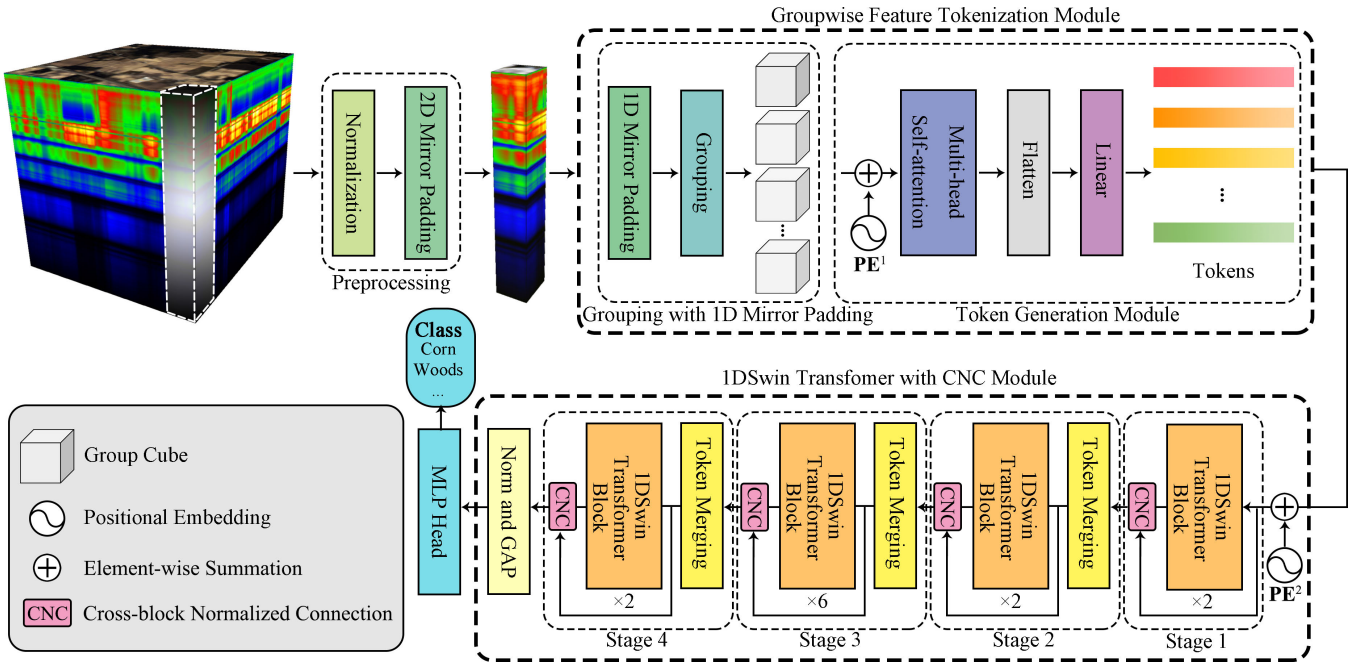


Fig. 1. Overview of our SS1DSwin. After the input HSI is processed by the preprocessing module, the image patch is cropped centered at the given pixel, which is fed to GFTM for generating token embeddings representing the local spatial–spectral relationship. Then, these token embeddings serve as the input of TCNCM followed by MLP to predict the label for the given pixel.

HSI as input and employed a multiscale local contrastive learning module to exploit multiscale semantic representations. Unfortunately, the extra pretraining cost of pretext tasks and the huge computational burden severely impede its deployment and application in real-world scenarios.

III. Methodology

The overall framework of SS1DSwin is depicted in Fig. 1. It is composed of three main components: preprocessing module, GFTM, and TCNCM. Considering an HSI $\mathbf{I} \in \mathbb{R}^{H\times W\times C}$, where H, W, and C are the height, width, and the number of spectral bands, respectively, we first recreate $\mathbf{I}$ as $\mathbf{I}^{\text{pn}} \in \mathbb{R}^{H'\times W'\times C}$ with normalization and padding. Then, the image patch $\mathbf{P} \in \mathbb{R}^{p\times p\times C}$ centered at the given pixel is cropped from $\mathbf{I}^{\text{pn}}$, which is fed into GFTM to output a series of token embeddings $\mathbf{T} = [\mathbf{T}_1, \mathbf{T}_2, \ldots, \mathbf{T}_C]$. Subsequently, $\mathbf{T}$ is passed to TCNCM followed by a multilayer perceptron (MLP) layer for the final predicted label $\mathbf{Y} \in \mathbb{R}^{1\times N}$, where N is the number of classes. For loss design, the commonly used cross entropy is utilized for training the proposed method. The description of notations used throughout this article is given in Nomenclature. The training pipeline of the proposed method is presented in Algorithm 1, and the detailed architecture can be found in Table I.

A. Preprocessing

In this section, we describe our preprocessing module, which aims to provide normalized and continuous results $\mathbf{I}^{\text{pn}}$ of the input $\mathbf{I}$. As channelwise normalization offers a better pixel value distribution compared with the normalization across all channels, we apply it to each channel of $\mathbf{I}$ and obtain the normalized result $\mathbf{I}^n \in \mathbb{R}^{H\times W\times C}$

$$\mathbf{I}_c^n = \text{Norm}(\mathbf{I}_c) = \frac{\mathbf{I}_c - \mathbf{I}_c^{\min}}{\mathbf{I}_c^{\max} - \mathbf{I}_c^{\min}}, \quad c = 1, 2, \ldots, C \tag{1}$$

where $\mathbf{I}_c$ and $\mathbf{I}_c^n$ denote the image matrices of $\mathbf{I}$ and $\mathbf{I}^n$ at the cth channel, respectively. $\mathbf{I}_c^{\max}$ and $\mathbf{I}_c^{\min}$ are, respectively, the maximum and minimum values of $\mathbf{I}_c$.

Instead of zero padding, as shown in Fig. 2(a), 2-D mirror padding with a padding size of $\lfloor (p/2) \rfloor$ is chosen to create the $\mathbf{I}^{\text{pn}}$ that preserves continuity at the edges of the image. Then, the corresponding image patch $\mathbf{P} \in \mathbb{R}^{p\times p\times C}$ is produced by cropping from $\mathbf{I}^{\text{pn}}$.

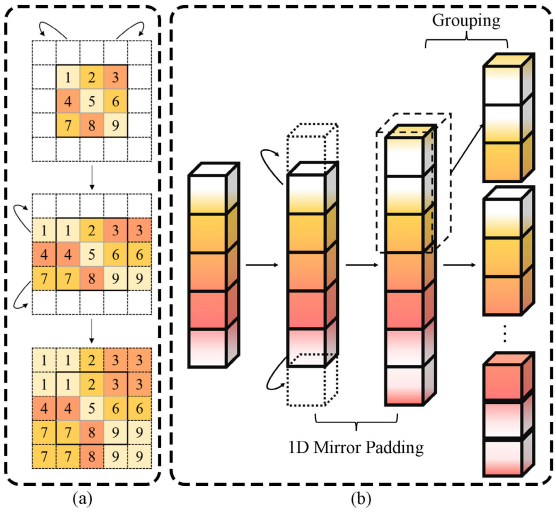


Fig. 2. (a) 2-D mirror padding. (b) Illustration of grouping with 1-D mirror padding.

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Algorithm 1 Training Procedure of SS1DSwin

Input: The input HSI $\mathbf{I} \in \mathbb{R}^{H\times W\times C}$.
Output: The predicted label $\mathbf{Y}$.

// Preprocessing
Compute $\mathbf{I}^{pn}$ using Eq. (1) and 2D mirror padding.
Crop $\mathbf{P}$ from $\mathbf{I}^{pn}$ with the given pixel at the center of $\mathbf{P}$.
for *epoch* from 1 to 3000 **do**
  // Groupwise Feature Tokenization Module (GFTM)
  Compute $\mathbf{P}^{pad}$ using Eq. (2).
  Compute $\mathbf{G}$ using Eq. (3).
  Compute $\mathbf{T}$ using Eq. (4), Eq. (5) and Eq. (6).
  // 1DSwin Transformer with CNC Module (TCNCM)
  $\mathbf{F}_{TM}^{s_1} = \mathbf{T}+\mathbf{PE}^2$.
  for i from 1 to 4 **do**
    if i != 1 **then**
      Compute $\mathbf{F}_{TM}^{s_i}$ using Eq. (7).
    end if
    Compute $\mathbf{F}_{b_{max_i}}^{s_i}$ using Eq. (8).
    Compute $\mathbf{F}_{CNC}^{s_i}$ using Eq. (11).
  end for
  Compute $\mathbf{F}^{gap}$ from $\mathbf{F}_{CNC}^{s_4}$.
  // Classification
  $\mathbf{Y} = MLP(\mathbf{F}^{gap})$.
  Update the model by optimizing the cross-entropy loss between $\mathbf{Y}$ and the ground-truth label.
end for

TABLE I
DETAILED ARCHITECTURE OF SS1DSWIN

Module	Block	Input size	Output size
GFTM	1D Mirror Padding	$p \times p \times C$	$p \times p \times C'$
	Grouping	$p \times p \times C'$	$C \times (p \times p) \times d_g$
	Multi-head Self-attention	$C \times (p \times p) \times d_g$	$C \times (p \times p) \times d_g$
	Flatten	$C \times (p \times p) \times d_g$	$C \times (p \times p \times d_g)$
	Linear	$C \times (p \times p \times d_g)$	$C \times 64$
TCNCM - Stage 1	1DSwin Transformer Block $\times 2$	$C \times 64$	$C \times 64$
	CNC	$C \times 64$ *and* $C \times 64$	$C \times 64$
TCNCM - Stage 2	Token Merging	$C \times 64$	$\frac{C}{2} \times 64$
	1DSwin Transformer Block $\times 2$	$\frac{C}{2} \times 64$	$\frac{C}{2} \times 64$
	CNC	$\frac{C}{2} \times 64$ *and* $\frac{C}{2} \times 64$	$\frac{C}{2} \times 64$
TCNCM - Stage 3	Token Merging	$\frac{C}{2} \times 64$	$\frac{C}{4} \times 64$
	1DSwin Transformer Block $\times 6$	$\frac{C}{4} \times 64$	$\frac{C}{4} \times 64$
	CNC	$\frac{C}{4} \times 64$ *and* $\frac{C}{4} \times 64$	$\frac{C}{4} \times 64$
TCNCM - Stage 4	Token Merging	$\frac{C}{4} \times 64$	$\frac{C}{8} \times 64$
	1DSwin Transformer Block $\times 2$	$\frac{C}{8} \times 64$	$\frac{C}{8} \times 64$
	CNC	$\frac{C}{8} \times 64$ *and* $\frac{C}{8} \times 64$	$\frac{C}{8} \times 64$
TCNCM	Norm and GAP	$\frac{C}{8} \times 64$	1×64
Classification	MLP Head	1×64	$1 \times N$

B. Groupwise Feature Tokenization Module (GFTM)

The goal of GFTM is to explore the local spatial–spectral relationship along the spatial dimension. We first describe the procedure of grouping with a 1-D mirror padding module that turns the input $\mathbf{P}$ to C group cubes $\mathbf{G} = \{\mathbf{G}_c \in \mathbb{R}^{p\times p\times d_g} \mid c \in [1, \ldots, C]\}$. To achieve the above goal, we then transform $\mathbf{G}$ to a set of token embeddings $\mathbf{T} = [\mathbf{T}_1, \mathbf{T}_2, \ldots, \mathbf{T}_c, \ldots, \mathbf{T}_C]$ with the proposed TGM. The details of these two modules are listed below.

1) Grouping With 1-D Mirror Padding Module: As shown in Fig. 2(b), the grouping with the 1-D mirror padding module is designed to provide enough consecutive and different overlapping cubes. Let $\mathbf{P} = [\mathbf{P}_1, \mathbf{P}_2, \ldots, \mathbf{P}_c, \ldots, \mathbf{P}_C]$ be the input image patch, where $\mathbf{P}_c \in \mathbb{R}^{p\times p}$ is the image matrix of $\mathbf{P}$ at the cth channel. $\mathbf{P}^{\text{pad}} = [\mathbf{P}_1^{\text{pad}}, \mathbf{P}_2^{\text{pad}}, \ldots, \mathbf{P}_c^{\text{pad}}, \ldots, \mathbf{P}_{C'}^{\text{pad}}] \in \mathbb{R}^{p\times p\times C'}$ is obtained by expanding the two ends of $\mathbf{P}$ symmetrically along the spectral dimension

$$\mathbf{P}_c^{\text{pad}} = \begin{cases} \mathbf{P}_{\left\lfloor \frac{d_g}{2} \right\rfloor + 1 - c}, & 1 \le c \le \left\lfloor \frac{d_g}{2} \right\rfloor \\ \mathbf{P}_{c - \left\lfloor \frac{d_g}{2} \right\rfloor}, & 1 + \left\lfloor \frac{d_g}{2} \right\rfloor \le c \le C + \left\lfloor \frac{d_g}{2} \right\rfloor \\ \mathbf{P}_{\left\lfloor \frac{d_g}{2} \right\rfloor + 1 + 2C - c}, & C + \left\lfloor \frac{d_g}{2} \right\rfloor + 1 \le c \le C + d_g - 1 \end{cases} \tag{2}$$

where d_g is the number of channels per group cube and $C' = C + d_g - 1$. Following that, the overlapping group cube $\mathbf{G}_c$ is acquired as follows:

$$\mathbf{G}_c = \left[\underbrace{\mathbf{P}_c^{\text{pad}}, \mathbf{P}_{c+1}^{\text{pad}}, \ldots, \mathbf{P}_{c+d_g-1}^{\text{pad}}}_{d_g} \right]. \tag{3}$$

2) Token Generation Module (TGM): For $\mathbf{G}_c$, we reshape it into a 2-D array $\mathbf{X}_c \in \mathbb{R}^{(p\times p)\times d_g}$. As is known, incorporating explicit representations of position information is especially important for Transformers [51], [52]. Therefore, we combine the positional embedding with the input $\mathbf{X}_c$ to expose the positional information

$$\mathbf{X}_c' = \mathbf{X}_c + \mathbf{PE}^1 \tag{4}$$

where $\mathbf{PE}^1 \in \mathbb{R}^{(p\times p)\times d_g}$ is a standard learnable positional embedding. In order to control the mixing of information within each group cube for richer representations, we decide to apply MSA on $\mathbf{X}_c'$ to yield $\mathbf{Z}_c \in \mathbb{R}^{(p\times p)\times d_g}$

$$\mathbf{Z}_c = \text{MSA}(\mathbf{X}_c') = \text{Concat}(\text{head}_1, \ldots, \text{head}_h)\mathbf{W}^{\text{out}}$$

where

$$\text{head}_i = \text{Attention}\left(\mathbf{X}_c'\mathbf{W}_i^Q, \mathbf{X}_c'\mathbf{W}_i^K, \mathbf{X}_c'\mathbf{W}_i^V\right) \tag{5}$$

where the projections are parameter matrices $\mathbf{W}_i^Q \in \mathbb{R}^{d_g\times d_a}$, $\mathbf{W}_i^K \in \mathbb{R}^{d_g\times d_a}$, $\mathbf{W}_i^V \in \mathbb{R}^{d_g\times d_a}$, and $\mathbf{W}^{\text{out}} \in \mathbb{R}^{(d_a\times h)\times d_g}$, and Concat$(\cdot)$ denotes the concatenation operation. Concretely, the function Attention$(\cdot)$ is formulated as follows:

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^T}{\sqrt{d_a}}\right)\mathbf{V}. \tag{6}$$

As a result, $\mathbf{Z}_c$ is flattened followed by a linear layer to output the token embedding $\mathbf{T}_c \in \mathbb{R}^{d_t}$. By stacking all the token embeddings, the output of TGM is denoted as $\mathbf{T} \in \mathbb{R}^{C\times d_t}$.

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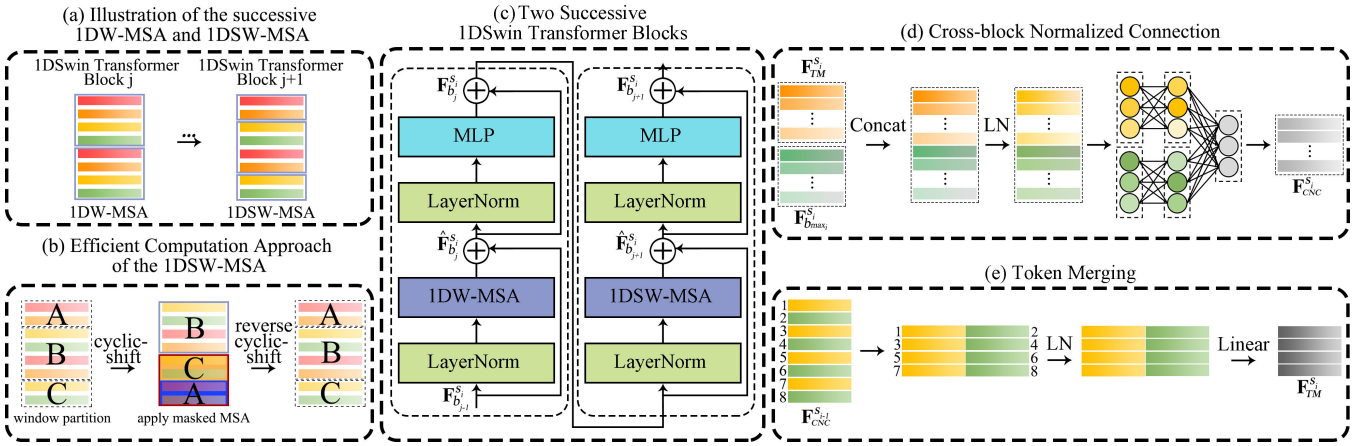


Fig. 3. (a) Two successive 1DW-MSA and 1DSW-MSA with different window partitioning strategies. (b) Pipeline of 1DSW-MSA. (c) Internal structure of two successive spectral 1DSwin Transformer blocks. (d) Process of handling the input token embeddings with CNC. (e) Diagram of a token merging layer.

C. 1DSwin Transformer With CNC Module (TCNCM)

To exploit comprehensive spatial–spectral representation, our proposed TCNCM aims to construct the hierarchical spatial–spectral relationship along the spectral dimension by 1DW-MSA and 1DSW-MSA and enhance the discriminative ability through CNC. TCNCM takes the sum of token embeddings $\mathbf{T}$ and a learnable positional embedding $\mathbf{PE}^2 \in \mathbb{R}^{C\times d_t}$ and fed it into the following four stages. After all these stages, layer normalization (LN) and global average pooling (GAP) are used to produce the final result $\mathbf{F}^{\text{gap}} \in \mathbb{R}^{1\times d_t}$ followed by an MLP layer to generate the predicted label $\mathbf{Y}$.

In regard to four stages, the input of stages 1–4 is different. In stage 1, the token merging layer is not required, and the input is $\mathbf{T} + \mathbf{PE}^2$ marked as $\mathbf{F}^{s_1}_{\text{TM}}$ for the sake of convenience. In stage i (i = 2–4), the input $\mathbf{F}^{s_{i-1}}_{\text{CNC}} = [\mathbf{F}^{s_{i-1}}_{\text{CNC}_1}, \mathbf{F}^{s_{i-1}}_{\text{CNC}_2}, \ldots, \mathbf{F}^{s_{i-1}}_{\text{CNC}_{(C/2^{i-1})}}] \in \mathbb{R}^{(C/2^{i-1}\times d_t)}$ is downsampled by the token merging layer as $\mathbf{F}^{s_i}_{\text{TM}} \in \mathbb{R}^{(C/2^i\times d_t)}$. In each stage, a stack of 1DSwin Transformer blocks are employed to produce hierarchical representations $\mathbf{F}^{s_i}_{b_j} \in \mathbb{R}^{(C/2^i\times d_t)} (b_1 \le b_j \le b_{\max_i})$. In addition, CNC is designed to connect and adaptively fuse $\mathbf{F}^{s_i}_{\text{TM}}$ and $\mathbf{F}^{s_i}_{b_{\max_i}}$ to obtain $\mathbf{F}^{s_i}_{\text{CNC}} \in \mathbb{R}^{(C/2^i\times d_t)}$. As the core component of each stage, each 1DSwin Transformer block consists of two LN layers, a 1DW-MSA (1DSW-MSA), residual connections, and a two-layer MLP. In the following, the details of abovementioned models will be listed.

1) Token Merging: The token merging layer aims to reduce the number of tokens as the network gets deeper for hierarchical representation as well as low computation cost. In stage i (i = 2–4) illustrated in Fig. 3(e), the token merging layer concatenates odd and even indexed elements of $\mathbf{F}^{s_{i-1}}_{\text{CNC}}$ alternatively in depth and applies LN and linear layers on the concatenated features $\mathbb{R}^{(C/2^{i+1}\times 2d_t)}$, resulting in the downsampled feature $\mathbf{F}^{s_i}_{\text{TM}}$ as the input of successive 1DSwin Transformer blocks. The above procedure is formally expressed in the following equation:

$$\mathbf{F}^{s_i}_{\text{TM}} = \text{LN}\left(\left[\text{Concat}\left(\mathbf{F}^{s_{(i-1)}}_{\text{CNC}_{2m+1}}, \mathbf{F}^{s_{(i-1)}}_{\text{CNC}_{2(m+1)}}\right]\right)\mathbf{W}^{\text{TM}}, \quad 0 \le m \le \frac{C}{2^i} \tag{7}$$

where the values of $\mathbf{W}^{\text{TM}} \in \mathbb{R}^{2d_t\times d_t}$ are the learnable weights of the linear layer.

2) 1DSwin Transformer Block: Derived from shifted attention appearing in Swin Transformer [43], we introduce 1DW-MSA and 1DSW-MSA to build 1DSwin Transformer blocks with better representation as well as greater efficiency. Fig. 3(c) shows the details of two successive 1DSwin Transformer blocks. They are built by replacing the standard W-MSA and SW-MSA in two successive Swin Transformer blocks with our proposed 1DW-MSA and 1DSW-MSA, which can be formulated as follows:

$$\begin{aligned}
\hat{\mathbf{F}}^{s_i}_{b_j} &= \text{1DW-MSA}\left(\text{LN}\left(\mathbf{F}^{s_i}_{b_{j-1}}\right)\right) + \mathbf{F}^{s_i}_{b_{j-1}} \\
\mathbf{F}^{s_i}_{b_j} &= \text{MLP}\left(\text{LN}\left(\hat{\mathbf{F}}^{s_i}_{b_j}\right)\right) + \hat{\mathbf{F}}^{s_i}_{b_j} \\
\hat{\mathbf{F}}^{s_i}_{b_{j+1}} &= \text{1DSW-MSA}\left(\text{LN}\left(\mathbf{F}^{s_i}_{b_j}\right)\right) + \mathbf{F}^{s_i}_{b_j} \\
\mathbf{F}^{s_i}_{b_{j+1}} &= \text{MLP}\left(\text{LN}\left(\hat{\mathbf{F}}^{s_i}_{b_{j+1}}\right)\right) + \hat{\mathbf{F}}^{s_i}_{b_{j+1}}
\end{aligned} \tag{8}$$

where $\mathbf{F}^{s_i}_{b_j}$ is the output of the jth 1DSwin Transformer block in stage i (i = 1–4). As depicted in Fig. 3(a), 1DW-MSA adopts a regular window partitioning strategy that evenly splits the input sequential tokens into nonoverlapping 1-D windows, and each window containing w token embeddings is processed by MSA. To improve the cross-window communication, 1DSW-MSA employs a shifted window partitioning scheme that displaces the windows by $\lfloor (w/2) \rfloor$ token embeddings from the regular partitioned 1-D windows. In stage i, there are $1 + (C/2^{(i-1)}w)$ 1-D shifted windows that utilize 1DSW-MSA in the blocks, and the number of token embeddings in the kth 1-D shifted window is defined as follows:

$$\begin{cases} \left\lfloor \dfrac{w}{2} \right\rfloor, & k = 1 \\ w - \left\lfloor \dfrac{w}{2} \right\rfloor, & k = 1 + \dfrac{C}{2^{(i-1)}w} \\ w, & \text{otherwise} \end{cases} \tag{9}$$

where k $[1 \le k \le 1 + (C/2^{(i-1)}w)]$ denotes the index of 1-D shifted windows. Regarding the consecutive 1DSwin Transformer blocks, regular window partitioning and shifted window partitioning are alternatively applied so as to improve

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modeling ability efficiently. Similar to the design in [43], as shown in Fig. 3(b), we introduce cyclic shifting toward the top direction and apply a masked MSA in the last 1-D window for efficient batch computation to guarantee the number of batched windows keeps the same as that of regular window partitioning.

Supposing that each 1-D window contains w token embeddings, the computational complexity of a global MSA module and a 1-D window based on C token embeddings are

$$\begin{aligned}\Omega(\text{MSA}) &= 4Cd_t^2 + 2C^2d_t \\ \Omega(\text{1DW-MSA}) &= 4Cd_t^2 + 2Cwd_t \end{aligned} \tag{10}$$

where the former is quadratic to the number of token embeddings C, while the latter is linear when w is fixed. This suggests that the 1DSW-MSA is more competent than MSA to handle a large number of 1-D data.

3) Cross-Block Normalized Connection (CNC): The motivation of CNC is to connect the input $\mathbf{F}_{\text{TM}}^{s_i}$ and output $\mathbf{F}_{b_{\max_i}}^{s_i}$ of 1DSwin Transformer blocks in stage i ($i = 1$–4) to alleviate severe gradient explosion and facilitate better training. In Fig. 3(d), CNC is composed of a concatenated layer, an LN layer, and three linear layers. It connects the features before and after the 1DSwin Transformer blocks in each stage. In detail, the concatenated layer concatenates $\mathbf{F}_{\text{TM}}^{s_i}$ and $\mathbf{F}_{b_{\max_i}}^{s_i}$ and feeds the concatenated feature into the LN layer. In this way, the concatenation and LN layers normalize the distributions of the intermediate layers, enabling smoother gradients, faster training, and better generalization. Following the concatenated normalization layer, the output of the LN layer is divided equally into $\mathbf{F}_{\text{TM}}^{s_i'}$ and $\mathbf{F}_{b_{\max_i}}^{s_i'}$, which are then adaptively fused with three linear layers as the output $\mathbf{F}_{\text{CNC}}^{s_i}$ of stage i:

$$\mathbf{F}_{\text{CNC}}^{s_i} = \text{Concat}\left(\mathbf{F}_{\text{TM}}^{s_i'}\mathbf{W}^{c1}, \mathbf{F}_{b_{\max_i}}^{s_i'}\mathbf{W}^{c2}\right)\mathbf{W}^{c3} \tag{11}$$

where $\mathbf{W}^{c1}, \mathbf{W}^{c2} \in \mathbb{R}^{d_t \times d_t}$, and $\mathbf{W}^{c3} \in \mathbb{R}^{2d_t \times d_t}$ are the learnable weights of the three linear layers, respectively.

IV. Experiments

In this section, we describe the datasets in Section IV-A, followed by the evaluation metrics, comparative approaches, and implementation details in Section IV-B. For evaluating our SS1DSwin, we present and discuss quantitative and qualitative results in Section IV-C. Furthermore, we analyze the computational complexity of different approaches in Section IV-D and provide ablation studies in Section IV-E.

A. Datasets

1) Indian Pines Dataset: The Indian Pines dataset is collected using the Airborne Visible/Infrared Imaging Spectrometer (AVIRIS) sensor over Northwestern Indiana, USA, in 1992. It includes 145 × 145 pixels and 224 spectral reflectance bands ranging in wavelength from 0.4×10^{-6} to 2.5×10^{-6} m and is reduced to 200 bands by removing the bands covering the absorption region [104–108], [150–163], and [220–224]. Its landscape contains two-thirds crops and one-third forest or other natural perennial vegetation. The land cover in the dataset is categorized into 16 classes, such as *Alfalfa*, *Soybean Mintill*, and *Corn Mintill*.

2) Pavia University Dataset: The Pavia University dataset is provided by the Telecommunications and Remote Sensing Laboratory, Pavia University. It is gathered with reflective optics system imaging spectrometer (ROSIS) sensor during a flight campaign over Pavia, northern Italy. The image consists of 610 × 340 pixels with 103 spectral bands and is divided into nine classes, including *Asphalt*, *Meadows*, *Gravel*, *Trees*, *Metal Sheet*, *Bare Soil*, *Bitumen*, *Brick*, and *Shadow*.

3) Houston2013 Dataset: The Houston2013 dataset is acquired by NSF-funded Airborne Laser Mapping Center (NCALM) in June 2012 at the University of Houston campus and adjacent urban areas. The hyperspectral imagery consists of 144 spectral bands in the 380–1050 nm. The image has a resolution of 349 × 1905 and covers 15 classes, such as *Healthy Grass*, *Tree*, and *Highway*.

4) Salinas Scene Dataset: The Salinas Scene dataset is collected by the 224-band AVIRIS sensor over Salinas Valley, CA, and is characterized by high spatial resolution (3.7-m pixels). This dataset contains 16 classes, such as *Fallow*, *Stubble*, and *Celery*.

The distributions of training samples and testing samples in these four datasets are given in Table II. For each class of Salinas Scene dataset, 67 training samples are randomly selected as the training set, and the rest are the testing set.

B. Experimental Setup

1) Evaluation Metrics: In order to evaluate our method on four datasets comprehensively and fairly, we adopt overall accuracy (OA), average accuracy (AA), and Kappa coefficient (κ) to make quantitative analysis of different methods. Notably, we report the results over accuracy for each category in the four datasets.

2) Comparative Approaches: Our comparative experiments are carried out on traditional methods, classic deep learning-based methods, and Transformer-based methods, including KNN [7], SVM [6], RF [8], 1-D CNN [13], 2-D CNN [14], RNN [45], miniGCN [29], end-to-end fusion networks (FuNet) [29], hybrid spectral CNN (HybridSN) [19], vision transformer (ViT) [50], hyperspectral image bidirectional encoder representations from transformers (HSI-BERT) (HB) [35], SF [36], HiT [37], SSFTT [38], GAHT [39], LF [40], MFT [41], and our proposed SS1DSwin.

1) For the 1-D CNN, it consists of a 1-D convolution filter, a batch normalization (BN) layer followed by a rectified linear unit (ReLU) layer, and softmax activation for HSI classification.
2) For the 2-D CNN, the HSI is sequentially fed into three 2-D convolution blocks to cope with HSI classification task. Each block includes a 2-D convolution layer, a BN layer, a ReLU layer, and a maximum pooling layer. The filter sizes of the 2-D convolution layer in each block are $3 \times 3 \times 32$, $3 \times 3 \times 64$, and $1 \times 1 \times 128$, respectively.
3) For the RNN, it is implemented by multilayer GRU with two recurrent layers, where the number of features in the hidden state is 128.
4) For the miniGCN, the network block comprises a BN layer and a graph convolution layer followed by a ReLU layer, where the graph convolution layer has 128 neuron units. More details can be found in [29].

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TABLE II
DISTRIBUTIONS OF TRAINING AND TESTING SAMPLES IN FOUR DATASETS

Class No.	Indian Pines			Pavia University			Houston 2013			Salinas Scene		
	Class Name	Training	Testing	Class Name	Training	Testing	Class Name	Training	Testing	Class Name	Training	Testing
1	Corn Notill	50	1384	Asphalt	548	6304	Healthy Grass	198	1053	Brocoli Green Weeds1	67	1942
2	Corn Mintill	50	784	Meadows	540	18146	Stressed Grass	190	1064	Brocoli Green Weeds2	67	3659
3	Corn	50	184	Gravel	392	1815	Synthetic Grass	192	505	Fallow	67	1909
4	Grass Pasture	50	447	Trees	524	2912	Tree	188	1056	Fallow Rough Plow	67	1327
5	Grass Trees	50	697	Metal Sheets	265	1113	Soil	186	1056	Fallow Smooth	67	2611
6	Hay Windrowed	50	439	Bare Soil	532	4572	Water	182	143	Stubble	67	3892
7	Soybean Notill	50	918	Bitumen	375	981	Residential	196	1072	Celery	67	3512
8	Soybean Mintill	50	2418	Bricks	514	3364	Commercial	191	1053	Grapes Untrained	67	11204
9	Soybean Clean	50	564	Shadows	231	795	Road	193	1059	Soil Vinyard Develop	67	6136
10	Wheat	50	162				Highway	191	1036	Corn Senesced Green Weeds	67	3211
11	Woods	50	1244				Railway	181	1054	Lettuce Romaine 4wk	67	1001
12	Buildings Grass Trees Drives	50	330				Parking Lot1	192	1041	Lettuce Romaine 5wk	67	1860
13	Stone Steel Towers	50	45				Parking Lot2	184	285	Lettuce Romaine 6wk	67	849
14	Alfalfa	15	39				Tennis Court	181	247	Lettuce Romaine 7wk	67	1003
15	Grass Pasture Mowed	15	11				Running Track	187	473	Vinyard Untrained	67	7201
16	Oats	15	5							Vinyard Vertical Trellis	67	1740
	Total	695	9671	Total	3921	40002	Total	2832	12197	Total	1072	53057

5) For the FuNet, it is designed by appending a fully connected (FC) fusion layer to the 2-D CNNs and miniGCNs.
6) For the HybridSN, it consists of three 3-D convolution layers, one 2-D convolution layers, and three FC layers.
7) For the ViT, the network structure is referenced to [50], where the length of each token embedding is set to 64, and the number of Transformer encoders is 5.
8) For the HB, a region containing the given pixel is selected and flattened into a pixel sequence, and then, it is transformed by a linear layer followed by positional embedding and a multilayer bidirectional Transformer encoder.
9) For the SF, the input token embeddings are fed to five cascaded Transformer encoder blocks. In each block, the number of heads in MSA is 4, and the MLP layer has a hidden dimension of 8.
10) For the HiT, it is a Transformer-based network that consists of a spectral-adaptive 3-D convolution projection module and convolution permutator for retrieving the subtle spatial–spectral discrepancies.
11) For the SSFTT, the 3-D patches derived from the input HSI are fed into spatial–spectral feature extraction, Gaussian-weighted feature tokenizer, and Transformer encoder followed by a softmax-based linear classifier to determine the label of each pixel.
12) For the GAHT, the input patches are processed with three stages in sequence, where each stage is composed of an embedding module using grouped convolutions, two Transformer blocks, and an LN layer.
13) For the LF, the original HSI is first fed into a two-layer CNN for dimension reduction. Then, a group convolutional block is employed for extracting patches, and simple linear iterative clustering (SLIC) is used for further extracting of superpixel features. Finally, the output features are passed to the local-enhanced Transformer encoder and a global Transformer encoder for final predicted label.
14) For the MFT, the HSI patch passes through a Conv3D layer and a HetConv2D layer. Both of these layers are followed by their respective BN and ReLU layers. Afterward, the output is flattened into feature patch tokens followed by Transformer encoder blocks for predicting class label.

3) Implementation Details: To be consistent with other experimental setups, our proposed SS1DSwin has four heads in the MSA layer ($h = 4$) and eight hidden dimensions in the MLP layer. We set d_t to 64 and d_a to 16. The number of 1DSwin Transformer blocks in four stages is set to 2, 2, 6, and 2, respectively. Following the work [36], Adam is used as the optimizer with a weight decay factor of 0.9 for every 300 epochs and a batch size of 64 during training. The initial learning rate of Adam is $2e^{-4}$. All the methods are trained on the device with an Intel Xeon E5-2620 v4 CPU and an NVIDIA GTX 1080 Ti 11-GB GPU.

C. Results and Discussion

1) Quantitative Analysis: We conduct extensive experiments on the Indian Pines dataset, Pavia University dataset, Houston2013 dataset, and Salinas Scene dataset over the metrics OA, AA, and κ, and then display the comparative results in Tables III–VI. From the results, we can clearly make the following observations. First, only with handcrafted features, traditional methods (KNN, RF, and SVM) struggle to classify the pixels accurately. Second, benefiting from the superior discriminative ability of convolution and GCNs, classic deep learning-based methods (1-D CNN, 2-D CNN, RNN, miniGCN, FuNet, and HybridSN) generally produce better results than traditional methods. Third, when it comes to Transformer-based methods, it is unexpected to observe that ViT just performs as well as 1-D CNN. To explain the reason, it may be attributed to the fact that pixelwise classifiers (1-D CNN and ViT) focus on the given pixel but ignore the impact brought by its neighbors, thereby degrading the classification accuracy obviously. In addition, HB and ViT only translate Transformers from NLP or CV to HSI classification directly and even produce similar performance with 2-D CNN and RNN. Fourth, although other methods (SF, HiT, SSFTT, GAHT, LF, and MFT) have greater potential on capturing long-range spatial–spectral features, they still fall behind ours by a large margin of 2.45%, 7.05%, 5.17%, 3.85%,

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TABLE III

QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE INDIAN PINES DATASET

Class No.	Conventional Classifiers			deep learning-based Networks						Transformer-Based Networks						
	KNN [7]	RF [8]	SVM [7]	1D CNN [13]	2D CNN [14]	RNN [46]	miniGCN [29]	FuNet [29]	HybridSN [19]	ViT [51]	SF [37]	HiT [38]	SSFTT [39]	GAHT [40]	MFT [42]	**SS1DSwin**
1	45.45	57.80	67.34	47.83	65.90	69.00	72.54	68.50	83.74	53.25	70.52	76.16	78.25	81.00	65.10	**83.89**
2	46.94	56.51	67.86	42.35	76.66	58.93	55.99	79.59	88.78	66.20	81.89	87.12	87.12	79.46	**95.54**	88.65
3	77.72	80.98	93.48	60.87	92.39	77.17	92.93	99.46	**100.00**	86.41	91.30	96.20	97.83	62.50	96.74	98.91
4	84.56	85.68	94.63	89.49	93.96	82.33	92.62	95.08	95.08	89.71	**95.53**	94.18	94.85	94.63	94.63	95.30
5	80.06	79.34	88.52	92.40	87.23	67.72	94.98	95.70	92.11	87.66	85.51	92.54	95.84	96.56	**99.86**	97.42
6	97.49	95.44	94.76	97.04	97.27	89.07	98.63	99.54	99.54	89.98	99.32	96.58	98.63	**100.00**	97.49	97.95
7	64.81	77.56	73.86	59.69	77.23	69.06	64.71	75.93	90.09	72.22	81.81	82.57	82.24	**95.75**	74.51	87.04
8	48.68	58.85	52.07	65.38	57.03	63.56	68.78	68.90	79.69	66.00	75.48	67.95	71.55	84.66	**88.21**	83.13
9	44.33	62.23	72.70	**93.44**	72.87	65.07	69.33	71.63	84.57	57.09	73.76	82.62	78.72	60.82	82.98	85.28
10	96.30	95.06	98.77	99.38	**100.00**	95.06	98.77	99.38	98.77	97.53	98.77	**100.00**	**100.00**	99.38	**100.00**	**100.00**
11	74.28	88.75	86.17	84.00	92.85	88.67	87.78	89.55	89.07	87.62	93.17	94.37	94.69	97.43	97.99	**98.63**
12	15.45	54.24	71.82	86.06	88.18	50.00	50.00	91.52	**100.00**	63.94	78.48	91.21	89.39	82.12	96.67	97.58
13	91.11	97.78	95.56	91.11	**100.00**	97.78	**100.00**	**100.00**	**100.00**	95.56	**100.00**	**100.00**	97.78	**100.00**	**100.00**	**100.00**
14	33.33	56.41	82.05	84.62	84.62	66.67	48.72	94.87	**97.44**	79.49	79.49	87.18	**97.44**	89.74	94.87	92.31
15	81.82	81.82	90.91	**100.00**	**100.00**	81.82	72.73	**100.00**	**100.00**	90.91	**100.00**	**100.00**	**100.00**	**100.00**	81.82	**100.00**
16	40.00	**100.00**	**100.00**	80.00	**100.00**	**100.00**	80.00	**100.00**	**100.00**	80.00	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**	**100.00**
OA (%)	59.17	69.80	72.36	70.43	75.89	70.66	75.11	79.89	87.59	71.86	81.76	82.75	84.10	86.89	87.44	**89.66**
AA (%)	63.90	76.78	83.16	79.60	86.64	76.37	78.03	89.35	93.68	78.97	87.81	90.54	91.52	89.00	91.65	**94.13**
κ	0.5395	0.6591	0.6888	0.6642	0.7281	0.6673	0.7164	0.7716	0.8588	0.6804	0.7919	0.8046	0.8195	0.8494	0.8558	**0.8819**

TABLE IV

QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE PAVIA UNIVERSITY DATASET

Class No.	Conventional Classifiers			Deep Learning-Based Networks						Transformer-Based Networks						
	KNN [7]	RF [8]	SVM [7]	1D CNN [13]	2D CNN [14]	RNN [46]	miniGCN [29]	FuNet [29]	HybridSN [19]	ViT [51]	SF [37]	HiT [38]	SSFTT [39]	GAHT [40]	MFT [42]	**SS1DSwin**
1	73.86	79.81	74.22	88.90	80.98	84.01	96.35	**96.67**	79.00	71.51	82.73	82.58	86.09	91.56	75.06	89.99
2	64.31	54.90	52.79	58.81	81.70	66.95	89.43	**97.60**	60.87	76.82	94.03	90.09	72.18	95.67	89.85	94.81
3	55.10	46.34	65.45	73.11	67.99	58.46	**87.01**	84.49	83.64	46.39	73.66	64.02	75.54	56.58	81.60	84.85
4	94.95	**98.73**	97.42	82.07	97.36	97.70	94.26	89.95	67.58	96.39	93.75	98.56	90.38	95.67	95.50	91.93
5	99.19	99.01	99.46	99.46	99.64	99.10	99.82	99.64	**100.00**	99.19	99.28	99.10	99.73	90.30	99.82	99.64
6	65.16	75.94	93.48	97.92	97.59	83.18	43.12	90.56	98.71	83.18	90.75	77.84	**99.80**	87.38	99.10	95.14
7	84.30	78.70	87.87	88.07	82.47	83.08	90.96	78.27	**98.27**	83.08	87.56	80.94	90.83	96.84	88.58	88.07
8	84.10	90.22	89.39	88.14	**97.62**	89.63	77.42	71.73	83.74	89.63	95.81	95.87	94.98	95.90	94.38	91.41
9	98.36	97.99	**99.87**	**99.87**	95.60	96.48	87.27	98.04	86.29	96.48	94.21	94.84	95.97	97.61	93.84	91.45
OA (%)	70.53	69.67	70.82	75.50	86.05	77.13	79.79	92.20	74.01	76.99	91.07	87.55	82.62	92.24	89.32	**93.04**
AA (%)	79.68	80.18	84.44	86.26	88.99	84.29	85.07	89.66	84.23	80.22	90.20	87.09	89.50	89.72	90.86	**91.92**
κ	0.6268	0.6237	0.6423	0.6948	0.8187	0.7101	0.7367	0.8951	0.6755	0.7010	0.8805	0.8338	0.7784	0.8954	0.8592	**0.9068**

TABLE V

QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE HOUSTON2013 DATASET

Class No.	Conventional Classifiers			Deep Learning-Based Networks						Transformer-Based Networks									
	KNN [7]	RF [8]	SVM [7]	1D CNN [13]	2D CNN [14]	RNN [46]	miniGCN [29]	FuNet [29]	HybridSN [19]	ViT [51]	HB [36]	SF [37]	HiT [38]	SSFTT [39]	GAHT [40]	LF [41]	MFT [42]	**SS1DSwin**
1	83.19	83.38	83.00	87.27	85.09	82.34	**98.39**	85.75	81.77	82.81	83.00	81.86	81.29	82.62	82.72	81.31	83.10	86.80
2	95.68	98.40	98.40	98.21	99.91	94.27	92.11	99.44	98.50	96.62	83.36	**100.00**	99.62	99.15	99.81	98.67	99.81	96.62
3	99.41	98.02	99.60	**100.00**	77.23	99.60	99.60	80.79	99.01	99.80	91.09	95.25	86.73	96.04	95.64	**100.00**	99.41	99.60
4	97.92	97.54	98.48	92.99	97.73	97.54	96.78	98.58	91.10	99.24	93.56	96.12	97.06	99.24	97.73	99.26	99.34	**99.43**
5	96.12	96.40	97.82	97.35	99.53	93.28	97.73	99.24	98.30	97.73	99.81	99.53	99.62	99.43	94.13	99.26	**100.00**	99.05
6	92.31	97.20	90.91	95.10	92.31	95.10	95.10	95.10	96.50	95.10	98.60	94.41	95.10	86.71	93.01	95.24	96.50	**98.60**
7	80.88	82.09	90.39	77.33	**92.16**	83.77	57.28	91.60	73.60	76.77	87.22	83.12	87.03	87.97	72.76	87.85	81.34	**87.78**
8	48.62	40.65	40.46	51.38	79.39	56.03	68.09	74.83	84.14	55.65	68.76	76.73	71.79	69.33	73.79	84.92	**86.13**	79.77
9	72.05	69.78	41.93	27.95	86.31	72.14	53.92	85.27	86.31	67.42	77.15	79.32	71.95	80.55	78.47	85.10	**86.59**	80.83
10	53.19	57.63	62.64	90.83	43.73	84.17	77.41	79.25	54.92	68.05	48.46	78.86	50.29	54.83	78.09	75.66	53.67	**92.08**
11	86.24	76.09	75.43	79.32	87.00	82.83	84.91	82.35	82.07	82.35	82.73	88.71	86.24	86.53	94.12	93.40	**98.01**	91.65
12	44.48	49.38	60.04	76.56	66.28	70.61	77.23	78.87	**92.80**	58.50	86.17	87.32	78.10	81.94	79.83	89.76	91.74	82.52
13	28.42	61.40	49.47	69.47	90.18	69.12	50.88	89.12	69.47	60.00	86.32	72.63	84.91	**94.74**	91.93	72.21	92.63	78.25
14	97.57	99.60	98.79	99.19	90.69	98.79	98.38	88.26	**100.00**	98.79	99.60	**100.00**	93.93	99.19	95.14	**100.00**	**100.00**	99.19
15	98.10	97.67	97.46	98.10	77.80	95.98	98.52	86.68	**100.00**	98.73	98.94	99.79	92.60	83.93	99.79	99.96	99.37	98.73
OA (%)	77.30	77.48	76.91	80.04	83.72	83.23	81.71	87.39	85.71	80.41	82.90	88.01	83.41	85.29	86.61	90.26	89.39	**90.46**
AA (%)	78.28	80.35	78.99	82.74	84.35	85.04	83.09	87.68	87.23	82.50	85.65	88.91	85.09	86.81	88.46	90.84	91.18	**91.39**
κ	0.7538	0.7564	0.7494	0.7835	0.8231	0.8183	0.8018	0.8631	0.844887	0.7876	0.8151	0.8699	0.8198	0.8404	0.8547	0.8942	0.8840	**0.8964**

0.20%, and 1.07% over OA in the Houston2013 dataset, respectively. The main reason is that 3-D attributes of HSI and image redundancy are not implicitly modeled with the insufficient training data.

As is well known, with the recognized areas of various shapes, these four datasets have their own challenges. Therefore, we take a closer look at those challenging classes that are easily misclassified in each dataset. Table III shows that there are several unbalanced classification results in the Indian Pines dataset. For example, the category "*Alfalfa*" has poor results in SF, because the sample size of this class is relatively small. This may be due to the weak inductive bias that has been widely used in MSA [42]. On the contrary, HiT, SSFTT, GAHT, and MFT introduce inherent inductive biases (translation equivalence and locality [50]) by employing convolutional layers as appropriate constraints to learn strong representations [42]. With respect to SS1DSwin, TGM locally mines features within group cubes, and 1DW-MSA calculates self-attention only within local windows rather than the global range, yielding relatively uniform and higher results from small-scale training samples. The categories "*Bare Soil*" and "*Meadows*" are spatially mixed in the Pavia University dataset. The classes "*Gravel*" and "*Bricks*" from the Pavia University dataset, the classes "*Highway*" and "*Railway*" from the Houston2013 dataset, and the classes "*Grapes Untrained*" and "*Vinyard Untrained*" from the Salinas Scene dataset present spectral responses with similar appearance, respectively. In the task of discriminating these categories, SS1DSwin behaves better than the comparative approaches. HB and LF are inferior to SS1DSwin by 43.62% (92.08→48.46) and 16.42%

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TABLE VI
QUANTITATIVE PERFORMANCE OF DIFFERENT METHODS ON THE SALINAS SCENE DATASET

	Conventional Classifiers			Deep Learning-Based Networks						Transformer-Based Networks						
Class No.	KNN [7]	RF [8]	SVM [6]	1-D CNN [13]	2-D CNN [14]	RNN [46]	miniGCN [29]	FuNet [29]	HybridSN [19]	ViT [51]	SF [37]	HiT [38]	SSFTT [39]	GAHT [40]	MFT [42]	**SS1DSwin**
1	86.76	88.65	88.35	99.18	**100.00**	99.54	99.54	99.74	**100.00**	99.69	98.87	99.43	**100.00**	**100.00**	**100.00**	**100.00**
2	88.33	89.56	89.16	99.78	99.97	99.56	99.97	**100.00**	91.17	60.43	98.74	99.89	99.97	99.95	**100.00**	99.97
3	80.01	84.26	86.08	98.59	**100.00**	98.53	97.17	98.53	99.79	92.09	99.95	**100.00**	99.74	99.21	**100.00**	99.79
4	89.31	88.02	88.31	99.40	**100.00**	99.77	99.17	**100.00**	99.25	99.70	99.40	99.92	99.92	99.92	99.92	96.99
5	84.76	85.70	86.30	98.09	**99.54**	97.70	98.54	99.39	98.51	97.59	99.39	99.50	99.20	96.44	**99.54**	98.77
6	89.29	88.96	88.86	99.82	**100.00**	99.54	99.97	99.97	**100.00**	99.46	99.97	**100.00**	**100.00**	99.90	**100.00**	**100.00**
7	88.77	87.87	89.30	99.72	**100.00**	97.81	99.60	99.83	99.91	99.20	99.12	99.63	**100.00**	**100.00**	**100.00**	99.94
8	62.36	71.88	74.45	**88.28**	72.09	63.03	59.63	72.30	81.64	66.75	71.89	75.21	77.36	78.11	78.64	87.67
9	87.23	88.31	88.73	99.77	99.98	96.59	99.85	99.23	99.89	98.73	99.32	99.79	99.53	97.70	**100.00**	99.90
10	72.00	74.41	78.80	90.84	94.58	90.28	94.05	94.74	98.44	91.59	96.85	97.32	97.79	**100.00**	99.53	95.76
11	78.00	78.00	79.40	99.20	99.50	96.40	97.20	99.30	98.70	95.60	98.70	98.90	98.90	87.71	**100.00**	98.00
12	88.22	84.12	88.01	99.84	**100.00**	99.19	**100.00**	99.73	99.73	97.74	**100.00**	99.89	99.89	**100.00**	99.84	**100.00**
13	86.68	87.55	87.66	99.41	**100.00**	94.35	99.18	**100.00**	99.06	94.11	99.88	**100.00**	99.65	99.88	**100.00**	**100.00**
14	79.53	81.87	81.21	95.41	98.70	92.02	96.91	97.01	98.80	94.82	95.31	98.80	97.41	95.81	97.91	**99.00**
15	50.88	49.55	51.72	49.59	61.77	54.84	79.21	71.50	**90.38**	60.42	87.61	76.07	75.41	77.79	79.45	89.27
16	80.24	83.34	85.17	99.31	98.97	98.05	99.14	99.14	96.78	97.01	98.39	98.62	98.62	**99.77**	98.91	99.37
OA (%)	75.69	78.02	79.52	89.73	88.50	84.28	87.88	89.66	93.82	83.26	91.71	91.16	91.51	91.56	92.57	**95.45**
AA (%)	80.77	82.00	83.22	94.76	95.32	92.33	94.95	95.65	97.00	90.31	96.46	96.44	96.46	95.76	97.11	**97.78**
κ	0.7340	0.7588	0.7750	0.8851	0.8718	0.8254	0.8656	0.8849	0.9312	0.8140	0.9079	0.9016	0.9055	0.9061	0.9172	**0.9493**

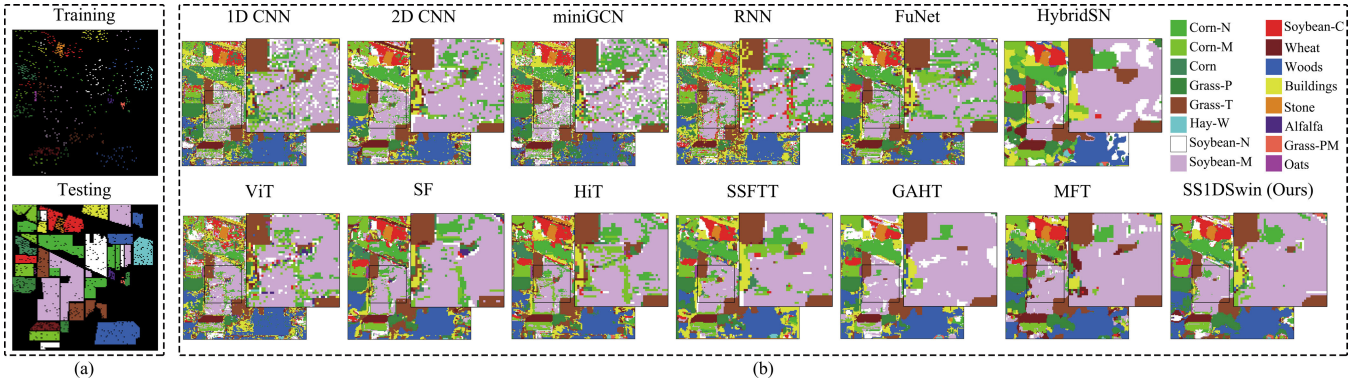


Fig. 4. Visualization results on the Indian Pines dataset. (a) Ground-truth maps of the training set and testing set. (b) Classification maps and 2.5 times zoomed ROIs of the compared methods. Corn-N: Corn Notill. Corn-M: Corn Mintill. Grass-P: Grass Pasture. Hay-W: Hay Windrowed. Soybean-N: Soybean Notill. Soybean-M: Soybean Mintill. Soybean-C: Soybean Clean. Buildings: Buildings Grass Trees Drives. Stone: Stone Steel Towers. Grass-PM: Grass Pasture Mowed.

(92.08→75.66) on recognizing class "*Highway*," respectively. As a consequence, our method has enough advantages to distinguish highly similar samples in the benchmark datasets with a higher result over three indicators.

2) Qualitative Analysis: To further emphasize the superiority of our method, we also show qualitative results along with zoomed regions of interest (ROIs) compared against other state-of-the-art methods in Figs. 4–7. Overall, without considering neighboring pixels, pixelwise classifiers (1-D CNN, RNN, miniGCN, and ViT) suffer from inevitable noise, yielding poorer results than patchwise ones (2-D CNN, FuNet, HybridSN, SF, HiT, SSFTT, GAHT, MFT, and SS1DSwin). As discussed in [53], 2-D CNN is subjected to severe over-smoothing problems at boundaries, which can be confirmed when we have a close-up view of "*Bricks*" in Fig. 5 and "*Commercial*" in Fig. 6. Limited by the CNN structure, HybridSN also suffers from over-smoothing problem at class "*Road*" and "*Running Track*" in Fig. 6. Faced with "*Bare Soil*" and "*Meadows*" in Fig. 5, SSFTT cannot figure them out with limited discriminative ability. Without the help of positional information, GAHT encounters over-smoothing issues particularly when dealing with "*Road*," "*Parking Lot2*," and "*Running Track*" in Fig. 6. When it comes to distinguishing the samples with highly similar appearances ("*Soybean Mintill*" and "*Corn Mintill*" in Fig. 4; "*Road*," "*Railway*," and "*Highway*" in Fig. 6; and "*Graspe Untrained*" and "*Vinyard Untrained*" in Fig. 7), SF, SSFTT, GAHT, HiT, and MFT perform helplessly due to the following factors: 1) by means of global attention [54], SF, SSFTT, GAHT, and MFT are prone to bring irrelevant parts inevitably when extracting representative features and 2) rather than global attention, HiT resorts to depthwise and pointwise convolution layers to build spatial–spectral features. Constrained by small receptive fields, it struggles to possess enough ability to differentiate similar samples. Different from the above methods, our proposed SS1DSwin can provide clearer details and less misclassification by sufficiently exploiting the jointly spatial–spectral features in different views with GFTM and TCNCM. Furthermore, we also present confusion matrices generated by SS1DSwin over all the four datasets in Fig. 8, respectively. As seen, only a few samples (*Gravel* and *Bricks* in the Pavia University dataset, and *Grapes Untrained* and *Vinyard Untrained* in the Salinas Scene dataset) are incorrectly classified, which proves the powerful discriminative ability of identifying different categories. Hence, the classification map of SS1DSwin shows more consistent agreement with the ground truth, which further indicates the superiority of the proposed methods on all four datasets.

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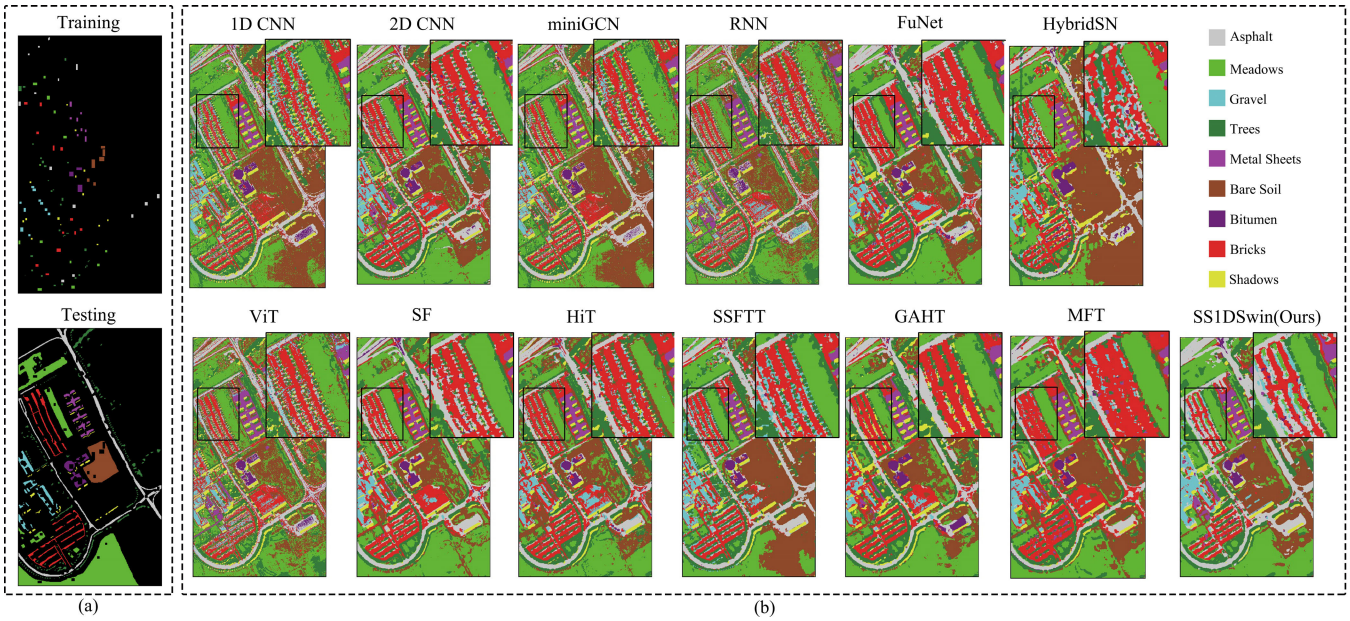


Fig. 5. Visualization results on the Pavia University Dataset. (a) Ground-truth maps of the training set and testing set. (b) Classification maps and two times zoomed ROIs of the compared methods.

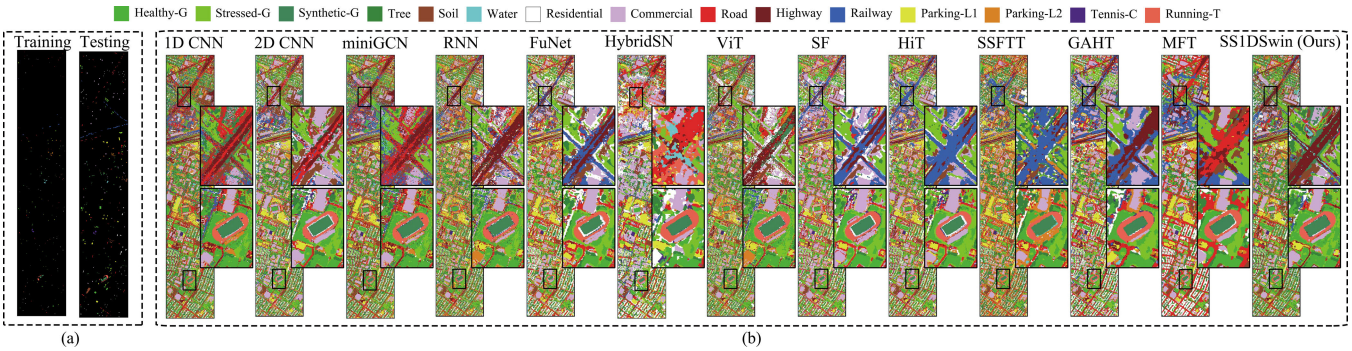


Fig. 6. Visualization results on the Houston2013 dataset. (a) Ground-truth maps of the training set and testing set. (b) Classification maps and four times zoomed ROIs of the compared methods. Healthy-G: Healthy Grass. Stressed-G: Stressed Grass. Synthetic-G: Synthetic Grass. Parking-L1: Parking Lot1. Parking-L2: Parking Lot2. Tennis-C: Tennis Court. Running-T: Running Track.

D. Computational Complexity Analysis

To illustrate the computation complexities of the proposed SS1DSwin and other methods, we report floating-point operations (FLOPs), the number of parameters (Params), and inference running time in Table VII. Similar to [55], we define the efficacy metric η to better evaluate the change among the boosted accuracy (OA), add-on FLOPs, and extra Params as follows:

$$\eta = \frac{\text{Norm}(\Delta\text{FLOPs}) + \text{Norm}(\Delta\text{Params}) + 1}{\Delta\text{OA} + 1} \tag{12}$$

where ΔOA is calculated in percentage format. η denotes the criterion that evaluates how many additional FLOPs and Params are required when the OA increases by 1%. Here, the operations of respective adding one to Norm(ΔFLOPs) + Norm(ΔParams) and ΔOA are to guarantee the numerator and denominator to be greater than zero. Apparently, the smaller the η is, the more efficient the model is. As observed in Table VII and Fig. 9, SS1DSwin requires relatively fewer FLOPs and Params among Transformer-based methods (ViT, SF, HiT, SSFTT, GAHT, MFT, and ours). As shown in Table VIII, it is worth noting that our proposed SS1DSwin achieves the lowest η when OA is taken into account. With regard to inference time, SS1DSwin elapses longer than SSFTT and GAHT yet with lower FLOPs and parameters. These results seem counterintuitive and could be attributed to the following facts: 1) in SSFTT, the 2-D and 3-D convolution layers in the hierarchical CNN module employ a series of 3×3 convolution, which is specially optimized and accelerated by the latest CUDA deep neural network (CUDNN) library [56], [57] and 2) compared with GAHT, several time-consuming operations ($\mathbf{PE}^1$ and $\mathbf{PE}^2$) and skip connections (CNC) that require fewer FLOPs and Params but more running time are adopted in our SS1DSwin. Consequently, our SS1DSwin is more appropriate in practical applications with respect to the balance of accuracy and computation cost.

E. Ablation Studies

1) Positional Embeddings: At the beginning of TGM and TCNCN, two positional embeddings $\mathbf{PE}^1$ and $\mathbf{PE}^2$ are introduced to give the order context to the nonrecurrent architecture

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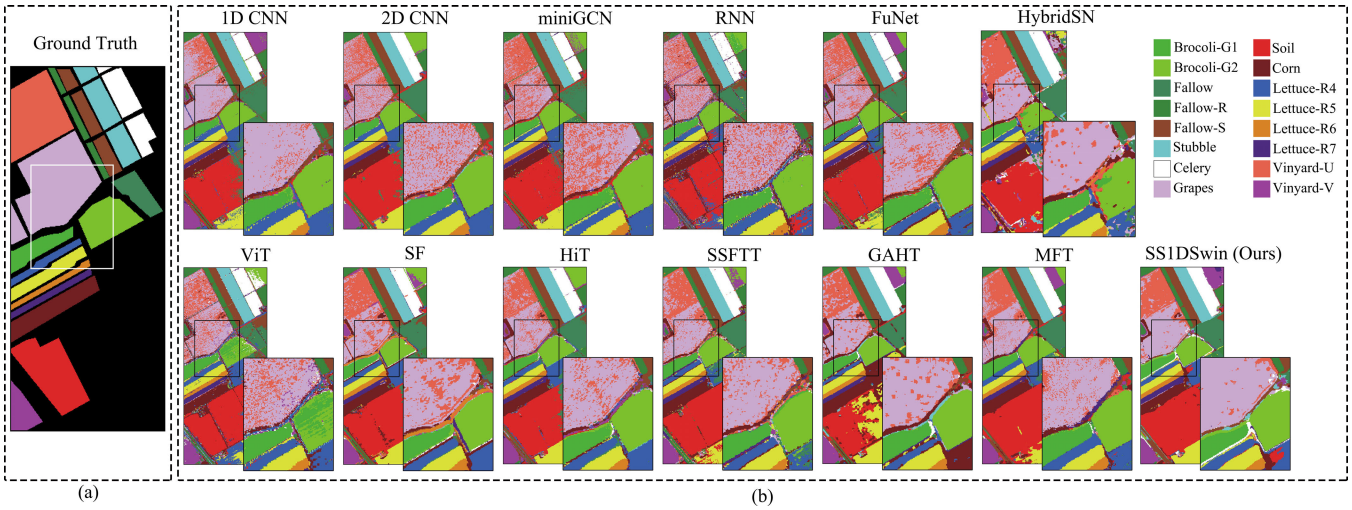


Fig. 7. Visualization results on the Salinas Scene dataset. (a) Ground-truth maps of the Salinas Scene dataset. (b) Classification maps and four times zoomed ROIs of the compared methods. Brocoli-G1: Brocoli Green Weeds1. Brocoli-G2: Brocoli Green Weeds2. Fallow-R: Fallow Rough Plow. Fallow-S: Fallow Smooth. Grapes: Grapes Untrained. Soil: Soil Vinyard Develop. Corn: Corn Senesced Green Weeds. Lettuce-R4: Lettuce Romaine 4wk. Lettuce-R5: Lettuce Romaine 5wk. Lettuce-R6: Lettuce Romaine 6wk. Lettuce-R7: Lettuce Romaine 7wk. Vinyard-U: Vinyard Untrained. Vinyard-V: Vinyard Vertical Trellis.

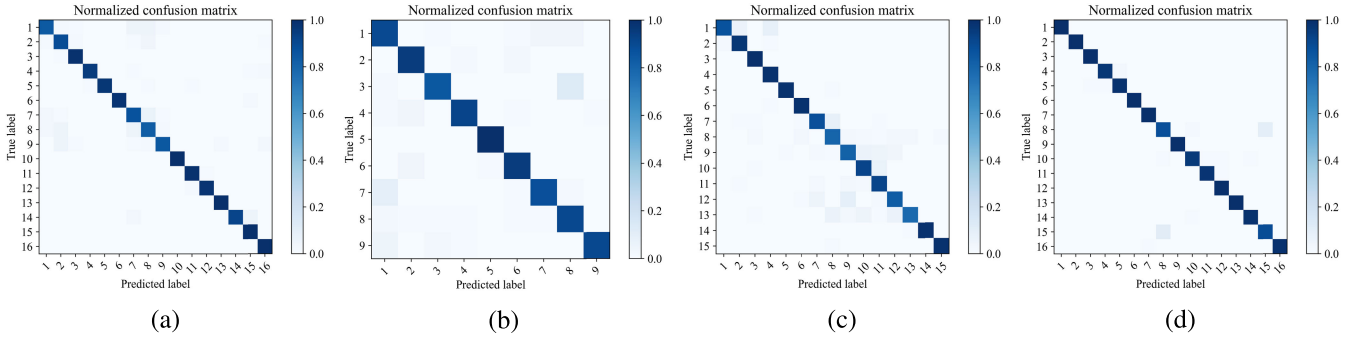


Fig. 8. Confusion matrices generated by SS1DSwin over four datasets. (a) Indian Pines. (b) Pavia University. (c) Houston2013. (d) Salinas Scene.

TABLE VII

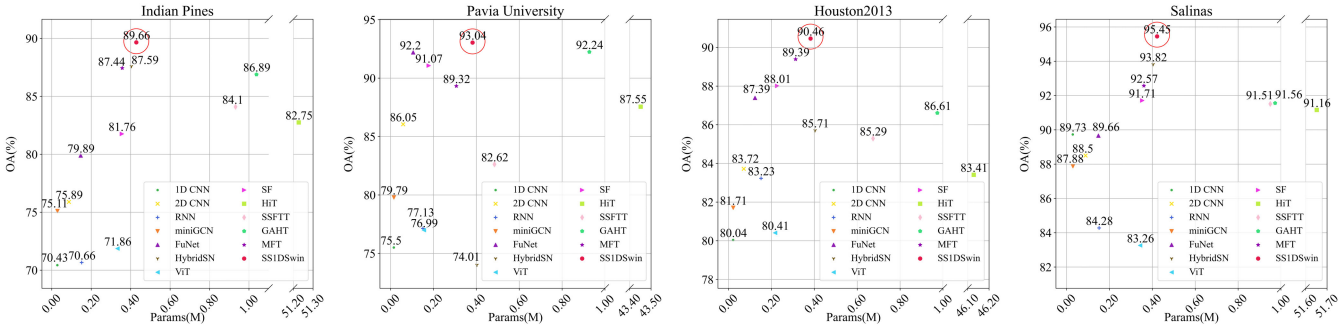
FLOPS, PARAMS, AND INFERENCE TIME OF COMPARATIVE METHODS ON FOUR DATASETS

Methods	Indian Pines			Pavia University			Houston2013			Salinas Scene		
	FLOPs(M)	Params(k)	Times (s)	FLOPs(M)	Params(k)	Times (s)	FLOPs(M)	Params(k)	Times (s)	FLOPs(M)	Params(k)	Times (s)
1D CNN [13]	**0.19**	**28.05**	**0.05**	**0.10**	**14.73**	**0.14**	**0.14**	**20.75**	**0.05**	**0.17**	**28.56**	**0.39**
2D CNN [14]	6.80	86.96	0.07	3.40	58.12	0.29	2.28	70.70	0.09	6.89	88.11	0.41
RNN [46]	61.56	151.44	0.73	30.78	150.54	1.63	21.51	151.31	0.66	60.94	151.44	0.92
miniGCN [29]	0.20	28.70	0.10	0.11	15.19	0.34	0.15	21.30	0.11	0.18	29.22	0.46
FuNet [29]	1.02	146.50	0.48	0.73	105.05	1.66	0.86	122.96	0.54	0.89	148.17	2.23
HybridSN [19]	12213.47	403.58	1.52	12211.64	402.68	5.90	12213.21	403.46	1.90	12213.47	403.58	7.51
ViT [51]	3.16	333.31	0.88	1.11	155.50	2.03	1.86	216.88	0.71	3.28	343.06	3.57
SF [37]	17.16	355.19	0.94	8.58	177.39	2.36	8.20	226.23	0.77	32.56	352.41	3.99
HiT [38]	251.80	51227.42	3.32	125.90	43451.71	13.82	70.34	46128.69	4.20	170.90	51652.06	20.43
SSFTT [39]	19.08	931.85	0.20	9.54	484.42	0.84	6.70	673.74	0.26	19.06	950.28	1.04
GAHT [40]	62.20	1038.16	0.45	31.10	927.11	1.81	17.81	973.58	0.55	7.08	972.62	2.89
MFT [42]	27.82	357.17	0.37	16.66	306.47	1.53	19.82	312.30	0.42	28.38	360.37	1.83
SS1DSwin	3.52	428.66	2.69	2.00	382.13	5.90	2.00	382.13	1.57	2.22	421.88	9.95

of multihead attention. In detail, $\mathbf{PE}^{\mathbf{1}}$ serves as a spatial positional identifier for the pixels in each group cube, and $\mathbf{PE}^{\mathbf{2}}$ describes the location of a token, so that each position is assigned to a unique representation. To investigate the effect of these two positional embeddings, we devise the following variants by removing them and present the experimental results in Table IX. As expected, our default model offers much better performance in conjunction with $\mathbf{PE}^{\mathbf{1}}$ and $\mathbf{PE}^{\mathbf{2}}$. After removing $\mathbf{PE}^{\mathbf{1}}$ and $\mathbf{PE}^{\mathbf{2}}$, there is an obvious performance drop by a large margin of 2.58% (93.04→90.46), 1.93% (91.92→89.99), and 0.0335 (0.9068→0.8733) over OA, AA, and κ, respectively. Compared with the model without $\mathbf{PE}^{\mathbf{1}}$ and $\mathbf{PE}^{\mathbf{2}}$, adding $\mathbf{PE}^{\mathbf{1}}$ or $\mathbf{PE}^{\mathbf{2}}$ will bring performance improvement over the three criteria. It demonstrates that positional embeddings help to provide the context of ordering when self-attention is utilized.

2) 1-D Mirror Padding: To verify the effectiveness of 1-D mirror padding, we consider the variants of SS1DSwin that use different padding strategies when constructing group cubes. More specifically, when extending $\mathbf{P} \in \mathbb{R}^{p\times p\times C}$ to $\mathbf{P}^{\text{pad}} \in \mathbb{R}^{p\times p\times C'}$ in spectral dimension, zero padding (13) sets the padding values with zeros, while circular padding [36] (14) implicitly sets them with the value of $\mathbf{P}_c$ periodically.

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(a) (b) (c) (d)

Fig. 9. Comparative OA and Params of all the methods over four datasets. The red circles represent our proposed SS1DSwin. (a) Indian Pines. (b) Pavia University. (c) Houston2013. (d) Salinas Scene.

TABLE VIII

EFFICACY METRIC η OF THE COMPARATIVE METHODS ON ALL DATASETS. ΔFLOPS, ΔPARAMS, AND ΔOA ARE COMPUTED BASED ON THE BASELINE (1-D CNN)

Methods	Indian Pines	Pavia University	Houston2013	Salinas Scene
	$\eta(\times 10^{-2})$	$\eta(\times 10^{-2})$	$\eta(\times 10^{-2})$	$\eta(\times 10^{-2})$
1D CNN [13]	100.00	100.00	100.00	100.00
2D CNN [14]	15.90	8.89	22.04	11.00
RNN [46]	101.33	47.41	31.20	25.92
miniGCN [29]	17.61	18.90	37.46	11.31
FuNet [29]	9.61	5.69	12.12	9.48
ViT [51]	41.88	40.61	75.09	24.27
SF [37]	8.71	6.46	12.48	9.44
HiT [38]	22.52	22.99	68.65	24.75
SSFTT [39]	7.45	13.37	17.72	9.05
GAHT [40]	7.25	7.14	16.81	8.46
MFT [42]	6.20	7.68	12.43	8.66
SS1DSwin	**5.05**	**5.52**	**9.07**	**6.21**

TABLE IX

ABLATION STUDIES ON POSITIONAL EMBEDDINGS $\mathbf{PE}^1$ AND $\mathbf{PE}^2$ FOR THE PAVIA UNIVERSITY DATASET

$\mathbf{PE}^1$	$\mathbf{PE}^2$	OA (%)	AA (%)	κ
×	×	90.46	89.99	0.8733
✓	×	90.95	90.52	0.8792
×	✓	90.84	89.02	0.8776
✓	✓	**93.04**	**91.92**	**0.9068**

TABLE X

ABLATION STUDY ON 1-D MIRROR PADDING FOR THE PAVIA UNIVERSITY DATASET

Padding Strategy	OA (%)	AA (%)	κ
Zero Padding	89.21	89.27	0.8571
Circular Padding	86.97	88.35	0.8284
1D Mirror Padding	**93.04**	**91.92**	**0.9068**

The experimental results are summarized in Table X. It is widely known that neighboring pixels have similar intensity pixels in most cases. By padding neighboring pixels rather than zeros, 1-D mirror padding brings 3.83%, 2.65%, and 0.0497 improvements over three criteria than zero padding, respectively. Note that the decrease brought by circular padding (6.07%, 3.57%, and 0.0784) is more significant than that caused by the zero padding (3.83%, 2.65%, and 0.0497). It is ascribed to the fact that circular padding introduces irrelevant and noisy values; thus, the classification accuracy drops obviously

$$P_c^{\text{pad}} = \begin{cases} 0, & 1 \le c \le \left\lfloor \frac{b}{2} \right\rfloor \\ P_{c-\left\lfloor \frac{b}{2} \right\rfloor}, & 1+\left\lfloor \frac{b}{2} \right\rfloor \le c \le C+\left\lfloor \frac{b}{2} \right\rfloor \\ 0, & C+\left\lfloor \frac{b}{2} \right\rfloor + 1 \le c \le C+b-1 \end{cases} \tag{13}$$

$$\mathbf{P}_c^{\text{pad}} = \begin{cases} \mathbf{P}_{c-\left\lfloor \frac{b}{2} \right\rfloor + C}, & 1 \le c \le \left\lfloor \frac{b}{2} \right\rfloor \\ \mathbf{P}_{c-\left\lfloor \frac{b}{2} \right\rfloor}, & 1+\left\lfloor \frac{b}{2} \right\rfloor \le c \le C+\left\lfloor \frac{b}{2} \right\rfloor \\ \mathbf{P}_{c-\left\lfloor \frac{b}{2} \right\rfloor - C}, & C+\left\lfloor \frac{b}{2} \right\rfloor + 1 \le c \le C+b-1. \end{cases} \tag{14}$$

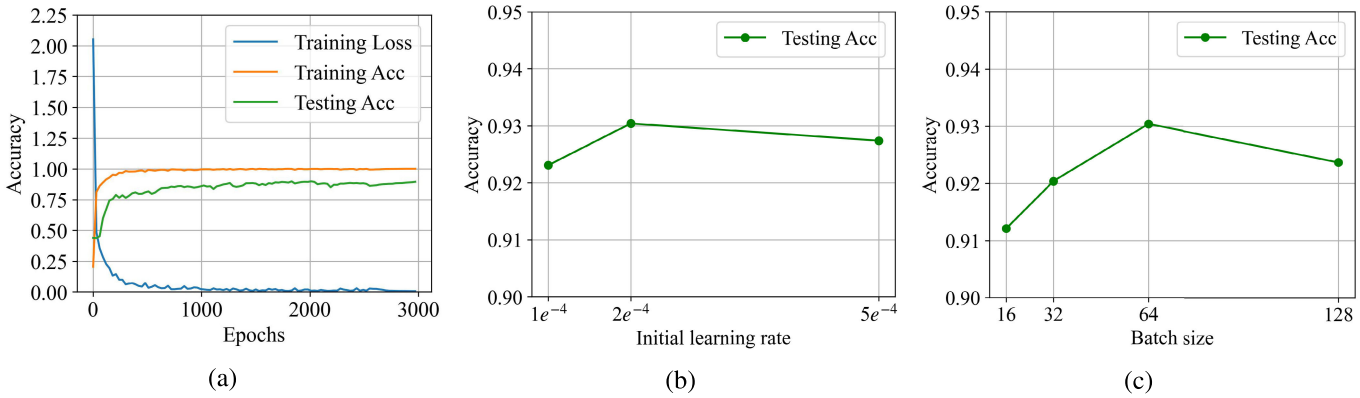
TABLE XI

ABLATION STUDY ON DOWNSAMPLING IN TCNCM FOR THE PAVIA UNIVERSITY DATASET

Downsampling Strategy	OA (%)	AA (%)	κ
Token Avg	91.59	89.90	0.8870
Token Max	90.02	89.08	0.8664
1D CNN	86.03	85.25	0.8118
Token Merging	**93.04**	**91.92**	**0.9068**

3) Downsampling in TCNCM: As the network gets deeper, the downsampling layer is always adopted to reduce the amount of computation as well as maintain the model's performance. To illustrate the advantage of token merging layers in TCNCM, we design another three alternatives for comparison in Table XI, including Token Avg, Token Max, and 1-D CNN. When involving Token Avg (Max), we average (maximize) the odd and even indexed elements of $\mathbf{F}_{\text{CNC}}^{s_{i-1}}$ alternatively in depth followed by an LN layer and a linear layer to generate $\mathbf{F}_{\text{TM}}^{s_i}$. In terms of 1-D CNN, it is applied to $\mathbf{F}_{\text{CNC}}^{s_{i-1}}$ with stride $= 2$ followed by an LN layer and a linear layer to downsample the output feature map $\mathbf{F}_{\text{TM}}^{s_i}$. As shown in Table XI, the models with Token Avg (Max) and 1-D CNN are evidently worse than ours. This implies that our token merging layer downsamples $\mathbf{F}_{\text{CNC}}^{s_{i-1}}$ efficiently without information loss by performing LN and linear layers on every two neighboring tokens, thus beating the runner-up (Token Avg) by 1.45% (93.04→91.59), 2.02%

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(a) (b) (c)

Fig. 10. Performance of SS1DSWin on the Pavia University dataset with the different epochs, initial learning rates, and batch sizes. (a) Training loss, training accuracy, and testing accuracy versus epochs. (b) Testing accuracy versus different initial learning rates. (c) Testing accuracy versus different batch sizes.

TABLE XII

ABLATION STUDY ON CNC FOR THE PAVIA UNIVERSITY DATASET. CON: CONCATENATION. LN: LAYER NORMALIZATION. LI: LINEAR LAYERS

Methods	CON	LN	LI	OA(%)	AA(%)	κ
SS1DSwin/CNC	–	–	–	89.33	90.47	0.8592
SS1DSwin/CLL	×	×	×	88.45	88.04	0.8447
SS1DSwin-L	×	✓	×	89.16	88.18	0.8538
SS1DSwin-CL	✓	✓	×	89.24	88.24	0.8563
SS1DSwin	✓	✓	✓	**93.04**	**91.92**	**0.9068**

(91.92→89.90), and 0.0198 (0.9068→0.8870) over OA, AA, and κ, respectively.

4) Cross-Block Normalized Connection (CNC): As a core component of TCNCM, CNC plays an important role in improving discriminative ability as well as better training. It contains a concatenated layer, an LN layer, and three linear layers. To evaluate the contribution of CNC, we build it up incrementally with the individual component and present ablation studies in Table XII. Specifically, in stage i ($i = 1$–4), SS1DSwin/concatenation, layer normalization and linear layers (CLL) directly treats the elementwise sum $\mathbf{F}_{\mathrm{TM}}^{s_i}$ of $\mathbf{F}_{b_{\max_i}}^{s_i}$ as $\mathbf{F}_{\mathrm{CNC}}^{s_i}$, which acts as the baseline for controlled experiments. By introducing the LN layer, $LN(\mathbf{F}_{\mathrm{TM}}^{s_i})$ and $LN(\mathbf{F}_{b_{\max_i}}^{s_i})$ are elementwise added to form $\mathbf{F}_{\mathrm{CNC}}^{s_i}$ shown as SS1DSwin-L. Different from SS1DSwin, the output of the LN layer is split in half, and these halves are elementwise added together, referred to as SS1DSwin-CL. In addition, we also list the performance of SS1DSwin/CNC by pruning away any residual link across all the 1DSwin Transformer blocks in each stage, which indicates that CNC is removed from the proposed model entirely. Compared with baseline (SS1DSwin/CLL), LN layers (SS1DSwin-L) can improve by 0.71% over OA (88.45→89.16), and the OA gap will be larger by 0.08% (89.16→89.24) if the concatenation is taken into account (SS1DSwin-CL). In contrast to SS1DSwin-CL, the OA of SS1DSwin is improved from 89.24 to 93.04 via adding linear layers. Therefore, the above observations and discussions underpin the necessity of these three components when constructing CNC.

5) Choices of Critical Parameters: To mitigate overfitting and increase the generalization capacity, the model should be trained and tested for an optimal number of epochs.

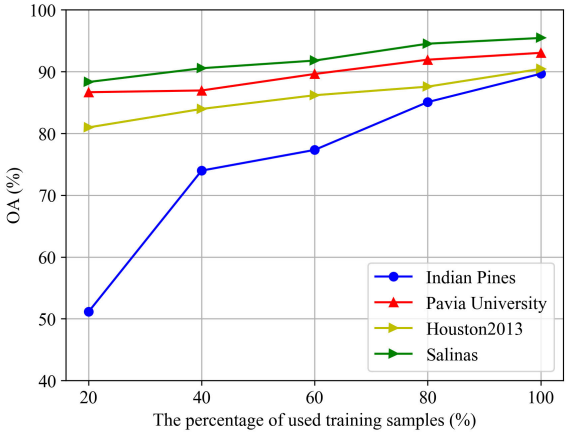


Fig. 11. Classification accuracy (OA) achieved by the proposed SS1DSwin with varying percentages of training samples on three benchmark datasets.

In Fig. 10(a), we monitor and portray the accuracy and loss convergence as epoch increases during training and testing. The convergence is achieved in approximately 500 epochs. To obtain more stable and accurate results, we set the number of epochs to 3000. As another two training parameters, initial learning rate and batch size play critical roles in outputting excellent deep models. As shown in Fig. 10(b) and (c), our model performs best on metric OA when the initial learning rate and batch size are set to $2e^{-4}$ and 64, respectively.

6) Percentage of Training Samples: It is generally known that the number of training samples directly affects the performance of Transformer-based methods. To check whether it works or not with the proposed SS1DSwin, we randomly evaluate the proportions of 20%, 40%, 60%, 80%, and 100% of the entire training dataset and depict the comparative results in Fig. 11. As expected, with the increase of training samples, the classification accuracy gradually improves. It is noted that OA tends to be stable when the percentage of training samples is greater than 80%. Furthermore, when the percentage of training samples in the Indian Pines dataset is less than 40%, the performance is unpleasant due to the insufficient samples for better training. Hence, it is reasonable to extrapolate that SS1DSwin possesses reliable stability in the task of HSI classification.

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V. Conclusion

In this article, we propose a novel Transformer-based model named SS1DSwin for HSI classification, which incorporates the properties of HSI with the network architecture of Swin Transformer. Specifically, SS1DSwin is competent in modeling local and hierarchical spatial–spectral relationships when constructing discriminative representations for better classification. SS1DSwin is mainly comprised of GFTM and TCNCM. Instead of applying global MSA to model long-range dependencies directly, GFTM reorganizes an image patch into overlapping group cubes and further generates groupwise token embeddings with MSA to learn the local spatial–spectral relationship along the spatial dimension. TCNCM extracts the hierarchical spatial–spectral relationship from these token embeddings with the help of 1DW-MSA and 1DSW-MSA and employs CNC to fuse features across 1DSwin Transformer blocks for improving discriminative ability as well as better training. Experiments on four HSI datasets of different scenarios have demonstrated that the proposed SS1DSwin achieves comparable results to the existing state-of-the-art HSI classifiers in terms of both classification accuracy and computation cost.

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